

# TECHNICAL REQUIREMENTS DOCUMENT

Sathvani AVC — Artificial Vocal Cord

Version 1.0 | August 2026 | Confidential

## 1. DOCUMENT CONTROL

| Field          | Details                                                                         |
|----------------|---------------------------------------------------------------------------------|
| Document Title | Technical Requirements Document — Sathvani AVC                                  |
| Version        | 1.0                                                                             |
| Date           | 26 August 2026                                                                  |
| Author         | Sathvani Engineering & AI Team                                                  |
| Classification | Confidential — Internal Use Only                                                |
| Stakeholders   | Founder/CEO, ML Lead, Embedded Engineer, Cloud Architect, Regulatory Consultant |

## 2. SYSTEM ARCHITECTURE OVERVIEW

The AVC system is a distributed edge-cloud architecture comprising four logical layers: (1) Wearable Edge (sensor acquisition + preprocessing), (2) Cloud Signal & Feature Layer (ingestion + feature engineering), (3) Cloud Inference Layer (deep learning phoneme reconstruction), and (4) Cloud Language & Synthesis Layer (sentence generation + TTS). All layers communicate over encrypted channels (TLS 1.3 / WPA3).

### Layer 1 — Wearable Edge:

PVDF Piezo (32-bit ADC) → MEMS Mic (I2S) → Pressure Sensor (I2C) → Airflow Sensor (I2C) → ESP32-S3 (sync + buffer) → WiFi/UDP streaming

### Layer 2 — Cloud Signal & Feature:

Ingestion Service → Feature Engineering (13 descriptors: RMS, SPL, Pressure, Velocity, Duration, Energy Ratio, Duration Norm, SPL/Vel Ratio, Pressure/RMS Ratio, ΔRMS, ΔSPL, ΔPressure, Phoneme-Class)

### Layer 3 — Cloud Inference:

Input Projection ( $D_{emb}=32$ ) → CNN ( $64 \rightarrow 128 \rightarrow 256$ , 1D conv, BatchNorm+GELU+Dropout) → BiLSTM ( $3 \times 256$  hidden, bidirectional) → 4-Head Self-Attention (residual+LayerNorm) → Classification Head (softmax, label smoothing)

### Layer 4 — Language & Synthesis:

CMU Dict Alignment → Edit-Distance Recovery → GPT-2 LM Re-ranking → FastSpeech2 + HiFi-GAN TTS → Personalized Voice Stream

## 3. HARDWARE TECHNICAL SPECIFICATIONS

### 3.1 Sensor Stack

| Sensor                                           | Interface           | Spec                                            | Role                                     | Status                  |
|--------------------------------------------------|---------------------|-------------------------------------------------|------------------------------------------|-------------------------|
| PVDF Piezoelectric Film                          | Analog → 32-bit ADC | Sensitivity ~50 pC/N; freq response 0.1Hz–20kHz | Primary signal (~60-70% SHAP importance) | Confirmed               |
| MEMS Microphone (SPH0625 or equiv)               | I2S                 | 16-bit, 16kHz, SNR 64dB                         | Acoustic capture; articulatory noise     | Confirmed on ESP32-S3   |
| Differential Pressure Sensor (MPXV7002 or equiv) | I2C                 | Range ±2kPa, 2.5% FS accuracy                   | Breath pressure during phonation         | Pending test            |
| Thermal Mass Airflow Sensor (S20 or equiv)       | I2C                 | Range 0–30 m/s, ±3% accuracy                    | Air velocity at mask outlet              | Integrated in prototype |

### 3.2 Edge Compute — ESP32-S3

| Parameter      | Specification                                           | Rationale                                                       |
|----------------|---------------------------------------------------------|-----------------------------------------------------------------|
| MCU            | ESP32-S3 (Xtensa LX7 dual-core @240MHz)                 | WiFi+BLE in single chip; <\$5 BOM; mature ecosystem             |
| RAM            | 512KB SRAM + 8MB PSRAM (external)                       | Multi-sensor buffer + WiFi stack                                |
| ADC            | 2× 12-bit SAR ADC (supplemented by external 32-bit)     | Piezo requires high dynamic range                               |
| Connectivity   | WiFi 4 (802.11 b/g/n) + BLE 5.0                         | UDP streaming to cloud; BLE for app pairing                     |
| Power          | USB-C 5V/1A; Li-Po 3.7V 2000mAh                         | ≥6hr continuous operation target                                |
| Firmware       | FreeRTOS + Arduino/ESP-IDF framework                    | Rapid iteration; I2S/I2C drivers validated                      |
| Sampling Rates | Piezo 1kHz   Mic 16kHz   Pressure 100Hz   Airflow 100Hz | 100Hz is quiet-compliant; async acquisition with timestamp sync |

## 4. DATA FLOW & PIPELINE SPECIFICATIONS

### 4.1 Edge-to-Cloud Streaming Protocol

**Protocol:** UDP over WiFi (unreliable but low-latency) with application-layer reliability.

**Packet Structure:** [Header: 8 bytes (seq\_no, timestamp\_ms, sensor\_mask)] [Payload: variable (interleaved sensor frames)] [Footer: 2 bytes CRC16]

**Rate:** ~50 KB/sec aggregate (piezo 1kHz×4B + mic 16kHz×2B + pressure 100Hz×4B + airflow 100Hz×4B)

**Buffering:** 500ms ring buffer on ESP32-S3; retransmission request via BLE if packet loss >2%

**Encryption:** DTLS 1.3 (datagram TLS) or AES-256-GCM pre-shared key rotation every 24hrs

### 4.2 Feature Engineering Pipeline (Cloud)

The 13-descriptor feature vector is computed per phoneme instance from raw sensor streams:

| Descriptor     | Formula / Computation                                  | Domain     | Rationale                         |
|----------------|--------------------------------------------------------|------------|-----------------------------------|
| RMS Amplitude  | $\sqrt{\text{mean}(x^2)}$ over phoneme window          | Energy     | Dominant predictor (SHAP 0.4134)  |
| SPL (dB)       | $20 \cdot \log_{10}(\text{RMS} / P_{\text{ref}})$      | Energy     | Sound pressure level              |
| Pressure (Pa)  | Raw differential pressure sensor                       | Pressure   | Subglottal/breath pressure        |
| Velocity (m/s) | Raw thermal airflow sensor                             | Flow       | Articulatory airflow velocity     |
| Duration (ms)  | End time – Start time                                  | Temporal   | Phoneme length discrimination     |
| Energy Ratio   | $\text{RMS} / (\mu_{\text{RMS}} + \epsilon)$           | Normalized | Relative energy within dialogue   |
| Duration Norm  | $\text{Duration} / (\mu_{\text{Duration}} + \epsilon)$ | Normalized | Relative duration within dialogue |

| Descriptor         | Formula / Computation                               | Domain       | Rationale                 |
|--------------------|-----------------------------------------------------|--------------|---------------------------|
| SPL/Velocity Ratio | $SPL / (Velocity + \epsilon)$                       | Ratio        | Pressure-flow coupling    |
| Pressure/RMS Ratio | $Pressure / (RMS + \epsilon)$                       | Ratio        | Pressure-energy coupling  |
| $\Delta RMS$       | $RMS_t - RMS_{t-1}$                                 | Differential | Local energy transition   |
| $\Delta SPL$       | $SPL_t - SPL_{t-1}$                                 | Differential | Local SPL transition      |
| $\Delta Pressure$  | $Pressure_t - Pressure_{t-1}$                       | Differential | Local pressure transition |
| Phoneme-Class      | Categorical encoding (vowel, stop, fricative, etc.) | Contextual   | Articulation group prior  |

**Preprocessing:** Stress markers stripped from phoneme labels; incomplete rows removed; temporal sorting by dialogue sequence preserved. Feature vector normalized per-dimension using z-score or min-max scaling before model input.

## 5. DEEP LEARNING ARCHITECTURE SPECIFICATIONS

### 5.1 Model Topology

| Layer               | Configuration                                             | Output Shape  | Parameters | Notes                        |
|---------------------|-----------------------------------------------------------|---------------|------------|------------------------------|
| Input               | 13 engineered descriptors                                 | (B, T, 13)    | —          | B=batch, T=seq_len           |
| Input Projection    | Linear(13 → 32) + GELU + LayerNorm                        | (B, T, 32)    | ~448       | Learnable embedding          |
| Conv Block 1        | Conv1d(32→64, k=3) + BN + GELU + Dropout(0.2)             | (B, T/2, 64)  | ~6K        | Local feature extraction     |
| Conv Block 2        | Conv1d(64→128, k=3) + BN + GELU + Dropout(0.2)            | (B, T/4, 128) | ~25K       | Hierarchical abstraction     |
| Conv Block 3        | Conv1d(128→256, k=3) + BN + GELU + Dropout(0.2)           | (B, T/8, 256) | ~98K       | High-level acoustic patterns |
| BiLSTM Layer 1      | BiLSTM(256 hidden, bidir)                                 | (B, T, 512)   | ~1.05M     | Forward+backward temporal    |
| BiLSTM Layer 2      | BiLSTM(512→256 hidden, bidir)                             | (B, T, 512)   | ~1.57M     | Deep temporal modeling       |
| BiLSTM Layer 3      | BiLSTM(512→256 hidden, bidir)                             | (B, T, 512)   | ~1.57M     | Contextual sequence          |
| Self-Attention      | 4 heads, d_k=128, residual + LayerNorm                    | (B, T, 512)   | ~1.05M     | Contextual refinement        |
| Classification Head | LayerNorm + FC(512→256) + GELU + Dropout(0.1) + FC(256→C) | (B, TFC)      | ~132K      | C = phoneme classes          |
| Output              | Softmax over C classes                                    | (B, T, C)     | —          | Probability distribution     |

**Total Trainable Parameters:** ~5.5M (lightweight; suitable for cloud inference and future edge quantization)

### 5.2 Training Configuration

| Hyperparameter | Value                                               | Justification                                           |
|----------------|-----------------------------------------------------|---------------------------------------------------------|
| Optimizer      | AdamW                                               | Decoupled weight decay; better generalization than Adam |
| Learning Rate  | 2×10 <sup>-3</sup>                                  | Aggressive initial LR for small dataset                 |
| Weight Decay   | 1×10 <sup>-4</sup>                                  | L2 regularization without gradient coupling             |
| Batch Size     | 32                                                  | Stable gradients; fits in single GPU                    |
| LR Schedule    | Cosine Annealing Warm Restarts                      | Periodic exploration; avoids premature convergence      |
| Loss Function  | Weighted Cross-Entropy + Label Smoothing (0.1)      | Handles class imbalance; reduces overconfidence         |
| Class Weights  | Inverse frequency                                   | Compensates for rare phonemes                           |
| Augmentation   | Gaussian noise (σ=0.04×mean_abs) + Time warping     | Effective dataset expansion from 48→1,920 samples       |
| Regularization | Dropout (0.2), BatchNorm, LayerNorm, Early Stopping | Multi-layer overfitting prevention                      |
| Hardware       | NVIDIA A10G / T4 or equivalent                      | Cloud GPU inference target                              |
| Framework      | PyTorch 2.x                                         | Dynamic graphs; strong research ecosystem               |

### 5.3 Explainability & Monitoring (Offline Path)

**SHAP + XGBoost Auxiliary Model:** Trained on identical 13-descriptor feature space. SHAP values computed per-feature using TreeSHAP. Primary use: offline feature importance monitoring, drift detection, and clinical

explainability (why did the model predict /p/ vs /b/?).

**Top SHAP-ranked features (research baseline):**

1. RMS Amplitude (0.4134) — dominant energy predictor
2. Phoneme-Class (0.2671) — articulation group prior
3.  $\Delta$ RMS (0.1842) — temporal energy transition
4. Duration (0.1336) — phoneme length
5. SPL (0.1058) — sound pressure level
6.  $\Delta$ SPL (0.0677) — temporal SPL transition

**Note:** Pressure, velocity, and ratio-based features showed near-zero SHAP contribution on the small research dataset. This is a **critical monitoring metric** — if these features remain uninformative on real patient data, the sensor stack should be re-evaluated.

## 6. LANGUAGE & SYNTHESIS LAYER

### 6.1 Phoneme-to-Word Reconstruction

**CMU Dictionary Alignment:** Predicted phoneme sequences  $P = \{p_1, p_2, \dots, p_n\}$  are matched against the CMU Pronouncing Dictionary using dynamic programming for optimal segmentation.

**Edit-Distance Recovery:** Levenshtein distance allows approximate matching when predicted phonemes deviate from dictionary entries (e.g., substitution of IH0→IH1 under noise).

**Candidate Generation:** For each phoneme subsequence, retrieve top-k dictionary matches (k=5 default). Compose candidate word sequences via beam search (beam width=3).

### 6.2 Language Model Re-Ranking

**Model:** GPT-2 (or lightweight open-source replacement e.g., LLaMA-3-1B, DistilGPT-2).

**Scoring:**  $\text{Score}(\text{sentence}) = -\text{Loss\_LM}(\text{sentence})$ . Lower perplexity = higher linguistic plausibility.

**Selection:** Top-1 candidate from beam search re-ranked by LM score.

**TR Decision — GPT-2 Replacement:** GPT-2 is non-commercial-friendly and computationally heavy.

**Recommendation:** Migrate to DistilGPT-2 or a fine-tuned Indic-LM (e.g., AI4Bharat IndicBERT) for Hindi/Telugu support. This reduces inference cost by ~60% and eliminates OpenAI licensing ambiguity.

### 6.3 Text-to-Speech Synthesis

| Component         | Technology                     | Spec                               | Rationale                                 |
|-------------------|--------------------------------|------------------------------------|-------------------------------------------|
| Acoustic Model    | FastSpeech2                    | Non-autoregressive; parallel mel-g | Low latency; controllable prosody         |
| Vocoder           | HiFi-GAN                       | GAN-based waveform synthesis       | Near-human naturalness; real-time capable |
| Speaker Embedding | d-vector / x-vector extraction | 256-dim speaker vector             | Personalization from 5-min pre-loss audio |
| Inference         | ONNX Runtime / TensorRT        | GPU batch inference                | <200ms TTS latency per utterance          |
| Output            | 16-bit PCM WAV, 22.05kHz       | Standard audio format              | Universal playback compatibility          |

## 7. CLOUD INFRASTRUCTURE & DEPLOYMENT

### 7.1 Cloud Platform Architecture

| Component           | Technology                               | Purpose                                 | Sizing (Year 1)                    |
|---------------------|------------------------------------------|-----------------------------------------|------------------------------------|
| API Gateway         | AWS API Gateway / Cloudflare             | Rate limiting, auth, routing            | 10K req/day                        |
| Ingestion Service   | AWS Kinesis / Apache Kafka               | Real-time sensor stream ingestion       | 50 KB/s per device × 2,300 devices |
| Feature Engineering | AWS Lambda / Python (Pandas, Numpy)      | Many-descriptor vector computation      | <100ms per utterance               |
| Inference Service   | AWS ECS (GPU) / PyTorch + ONNX Runtime   | ONNX-BERT M-Attention inference         | A10G instance; 100 req/sec         |
| TTS Service         | AWS ECS (GPU) / FastSpeech2 + HiFi-GAN   | Speech synthesis                        | Shared GPU pool with inference     |
| LM Re-Ranking       | AWS ECS (CPU) / HuggingFace Transformers | GPT-2/DistilGPT-2 scoring               | c6i.2xlarge; batch scoring         |
| Database            | PostgreSQL + Redis                       | User profiles, voice models, usage logs | 100GB storage; 1K IOPS             |

| Component      | Technology                        | Purpose                                     | Sizing (Year 1)          |
|----------------|-----------------------------------|---------------------------------------------|--------------------------|
| Object Storage | AWS S3                            | Audio recordings, model artifacts, datasets | 5TB encrypted            |
| Monitoring     | Prometheus + Grafana + CloudWatch | Latency, accuracy, drift, health            | Real-time dashboards     |
| CI/CD          | GitHub Actions + Terraform        | Automated testing, infra-as-code            | Full pipeline automation |

## 7.2 API Specifications

**POST /v1/ingest** — Stream raw sensor frames from device

Headers: Authorization: Bearer <token>, Content-Type: application/x-avc-stream

Body: Binary protobuf (seq\_no, timestamp, sensor\_mask, [frames])

Response: 202 Accepted {session\_id, estimated\_latency\_ms}

**POST /v1/synthesize** — Trigger phoneme reconstruction + TTS

Body: JSON {session\_id, user\_id, voice\_profile\_id, language\_code}

Response: 200 OK {audio\_url, phoneme\_sequence, confidence\_score, latency\_ms}

**GET /v1/health** — Service health check

Response: 200 OK {status, version, gpu\_utilization, queue\_depth}

**POST /v1/calibrate** — Per-user sensor calibration

Body: JSON {user\_id, baseline\_readings: {pressure, airflow, piezo\_rms}}

Response: 200 OK {calibration\_vector, confidence}

## 8. SECURITY & COMPLIANCE

| Domain                    | Requirement                        | Implementation                                             |
|---------------------------|------------------------------------|------------------------------------------------------------|
| Data Encryption (Transit) | TLS 1.3 / DTLS 1.3 for UDP         | Mandatory for all device-cloud and app-cloud communication |
| Data Encryption (Rest)    | AES-256-GCM                        | All patient audio, sensor data, and voice models           |
| Authentication            | OAuth 2.0 + JWT (RS256)            | Short-lived tokens (15min); refresh tokens (7 days)        |
| Authorization             | RBAC (Role-Based Access Control)   | Patient, Clinician, Admin, Researcher roles                |
| Audit Logging             | Immutable audit trail              | All data access logged to WORM storage (7 years)           |
| Anonymization             | k-anonymity + differential privacy | Research data export only after de-identification          |
| India Compliance          | DPDP Act 2023                      | Consent management; data localization; breach notification |
| US Compliance             | HIPAA                              | BAA with cloud provider; minimum necessary access          |
| EU Compliance             | GDPR                               | Right to erasure; data portability; DPO appointment        |
| Medical Device            | ISO 13485 + IEC 62304              | QMS for software lifecycle; risk management per ISO 14971  |

## 9. EDGE QUANTIZATION & FUTURE ROADMAP

**Current State:** Full inference runs in cloud. ESP32-S3 handles only acquisition and streaming.

**Phase 2 Edge Plan (2027):**

- Quantize CNN-BiLSTM to INT8 using ONNX Runtime + TensorFlow Lite Micro
- Target: partial phoneme classification on edge (top-3 candidates); full refinement in cloud
- Reduces cloud dependency for common phrases; enables offline mode
- Memory budget: ≤2MB model size for ESP32-S3 PSRAM

**Phase 3 Edge Plan (2028):**

- Evaluate ESP32-P4 or ARM Cortex-M55 + Ethos-U55 NPU for full edge inference
- Target: <100ms edge-only latency for 50 common phrases
- Cloud reserved for personalization, model updates, and analytics

## 10. TESTING & VALIDATION STRATEGY

**Unit Testing:** pytest for feature engineering, model forward pass, API endpoints

**Integration Testing:** End-to-end pipeline with synthetic sensor data (phantom speech simulator)

**Model Validation:** Stratified k-fold cross-validation (k=5); held-out patient cohorts; confusion matrix per phoneme class

**Latency Testing:** p50, p95, p99 latency benchmarks under load (Locust/k6)

**Security Testing:** OWASP ZAP penetration testing; fuzzing of ingestion API

**Clinical Validation:** 10-patient pilot at AIIMS; IRB-approved protocol; primary endpoint: intelligibility score (CIID)

## 11. OPEN TECHNICAL DECISIONS



- **LM Choice:** Stick with GPT-2 vs. migrate to DistilGPT-2 / Indic-LM? Decision needed by Phase 2 (Q1 2026).
- **Cloud Provider:** AWS vs. GCP vs. Azure? GCP offers better TPU pricing for inference; AWS has stronger India presence.
- **Edge Quantization:** Is INT8 BiLSTM feasible on ESP32-S3, or should we skip to ESP32-P4?
- **Database:** PostgreSQL sufficient, or move to time-series DB (TimescaleDB) for sensor data?
- **Monitoring:** Build custom SHAP drift detection, or use Evidently AI / WhyLabs?

## 12. REFERENCES

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